# Inspection Carte Mère : Slots DIMM Endommagés, Corrosion et Pins CPU

## 1. Contexte et Description Technique

Diagnostic des pannes situées côté carte mère : connecteurs DIMM fondus, accumulation de poussière conductrice, pistes coupées entre le socket CPU et la RAM, ou broches LGA tordues.

## 2. Symptômes Observés sur la Machine

Un canal mémoire entier (Canal A ou Canal B) reste inutilisable même avec une barrette de RAM neuve et testée fonctionnelle.

## 3. Protocole de Diagnostic et Résolution

1. Examiner l'intérieur du connecteur DIMM sous éclairage rasant pour détecter toute broche métallique pliée ou résidu de pâte thermique.

2. Nettoyer à la bombe à air sec et au nettoyant contact.

3. Si un canal complet est mort, démonter le processeur et inspecter les broches du socket CPU (LGA1700 / AM5) reliées au contrôleur mémoire intégré (IMC).

## 4. Articles et Retours d'Expérience Extraits du Web

**Extrait de [RAM & Motherboard Errors: Real Fixes That Work | Erropedia] :**
Quick answer: Pull every stick, test one at a time in slot A2, and if that boots but A2+the second stick in B2 doesn't, the problem is the slot, the CPU memory controller, or bent socket pins — not the RAM itself. What's actually happening here is that most 'RAM errors' people chase aren't RAM. They're the channel. A modern board runs dual channel across specific slot pairs (usually A2+B2 on a four-slot board, counting from the CPU). If one slot's traces are damaged, or a single LGA1700/AM5 pin under the socket is bent, one channel goes dark. The system either won't POST at all, throws a DRAM debug LED, or boots into single-channel mode with half the bandwidth you paid for. The stick is fine. The path to it isn't. I've seen this most often after a CPU cooler swap. You torque an NH-D15 down, the retention bracket flexes the board, and a pin that was borderline makes contact with nothing. Same story after a case move where the DIMMs unseat by a millimeter — DDR4 and DDR5 need a genuinely firm click on both latches, and 'it looks seated' is not the same as seated. Numbered fix steps - Clear CMOS and boot minimal. Pull the battery or short the CLR_CMOS jumper with the PSU unplugged, th...

**Extrait de [Dead RAM Slot? Fix It in 10 Minutes | Erropedia] :**
1. Bent CPU Socket Pins (The #1 Cause) I've seen this more times than I can count. Someone swaps a cooler, tightens it unevenly, and suddenly channel B or A2 stops working. The memory controller is on the CPU die. Bent pins under the socket break the connection to specific memory channels. You get a dead slot or a machine that won't post past 0x4C (or 0x50 on some ASUS boards). The beep pattern is usually three long, three short on older AM4 boards. How to verify Pop the CPU out. Grab a magnifying glass or your phone's macro lens. Look at the socket pins, especially near the corner where the memory traces run. You're looking for pins that are folded flat, tilted sideways, or obviously shorter than the rest. On LGA (Intel) sockets, look at the pads on the motherboard socket — bent pins are more subtle but still visible. The fix Don't bother with a mechanical pencil or tweezers if you're shaky. Use a fine-tipped sewing needle or a 0.5mm mechanical pencil with no lead. Gently nudge the pin back into position. Work slowly. If you snap a pin, you're boned — time for a new board or CPU. Re-seat the CPU and cooler with even pressure. Torque the cooler screws in a cross pattern. This fixes...

**Sources Web Associées :**

• RAM & Motherboard Errors: Real Fixes That Work | Erropedia (https://erropedia.com/solution.php?slug=dimm-slot-b2-dead-fixing-ram-and-motherboard-errors)

• Dead RAM Slot? Fix It in 10 Minutes | Erropedia (https://erropedia.com/solution.php?slug=ram-slot-not-working-fix-dead-dimm-slots-fast)

# Catalogue Visuel : 10 Photos Réelles (04_slot_motherboard_damage)

Ces 10 cas visuels issus du dataset illustrent les motifs caractéristiques de cette classe d'erreur :

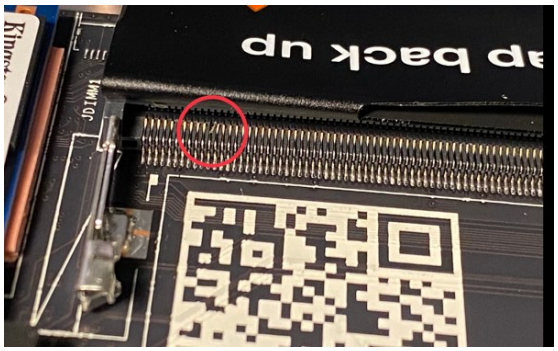


**Figure 01** — 04_slot_motherboard_damage (Échantillon #1)

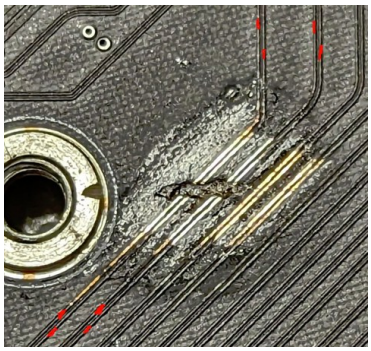


**Figure 02** — 04_slot_motherboard_damage (Échantillon #2)

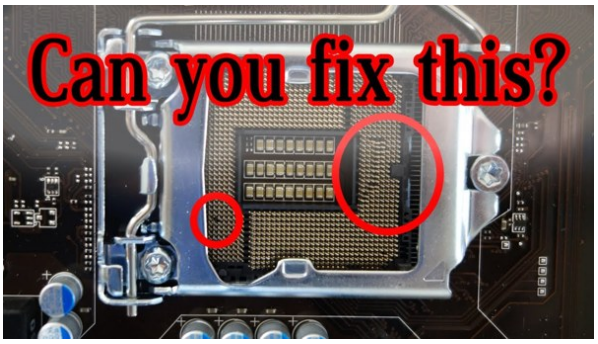



**Figure 03** — 04_slot_motherboard_damage (Échantillon #3)

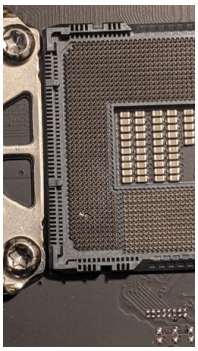


**Figure 04** — 04_slot_motherboard_damage (Échantillon #4)




**Figure 05** — 04_slot_motherboard_damage (Échantillon #5)

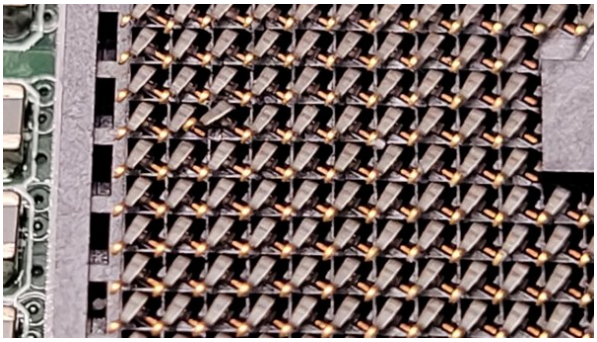


**Figure 06** — 04_slot_motherboard_damage (Échantillon #6)

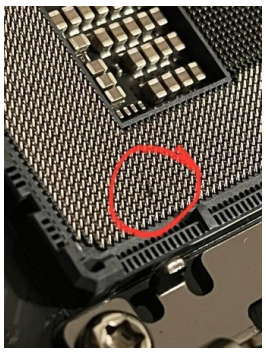


**Figure 07** — 04_slot_motherboard_damage (Échantillon #7)

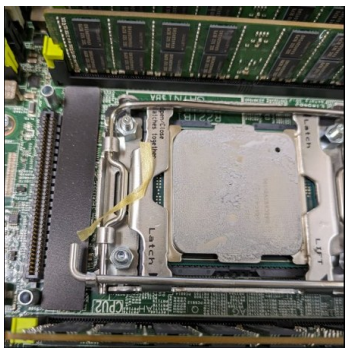


**Figure 08** — 04_slot_motherboard_damage (Échantillon #8)

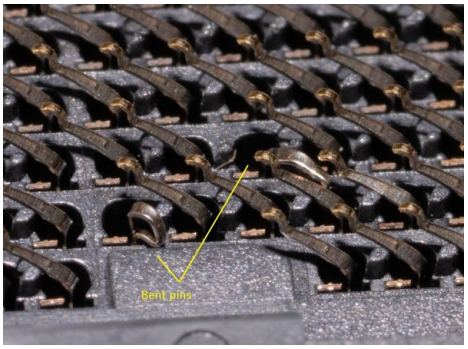



**Figure 09** — 04_slot_motherboard_damage (Échantillon #9)

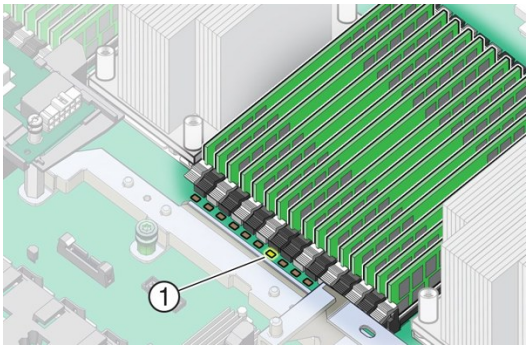



**Figure 10** — 04_slot_motherboard_damage (Échantillon #10)